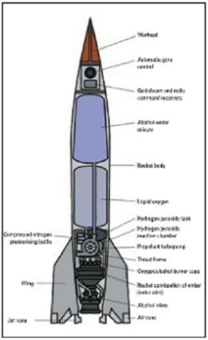

Figure 3: All modern rockets share the same concept as V-2, but instead of a warhead, the amateur rocket carries a useful payload

"And so LV2 ended up with a 32-bit processor running Debian Linux 2.4. Our launch tower became another Linux box, and we had a field server routing packets between the rocket, launch control and the launch tower... it was all fantastically fun. And for the first time, we really had enough horsepower in LV2 to consider starting active guidance," reveals Greenberg.

Open source components

Currently, to build and launch the rocket requires the combined efforts of several teams with their respective specialisation in airframes, avionics, communications, ground control, the payload, propulsion, software and uncertainty. As you can see, this is a mixture of software and hardware teams—and they all play important roles.

The airframe team is responsible for designing the LV rockets—how to construct an aluminium module, where to place motor casing, nozzles and coupler bulkhead, as well as the aeroshell construction and parachute recovery module, which hides inside a rocket and is used on its return to the ground.

The avionics team produces all the software [6] and deals with hardware modules (like the flight computer, discussed below). This team is responsible for reliable software interconnections during flight and after-flight analysis (via the syslog Linux-subsystem). For example, the team is deeply connected with the ground team in terms of radio-exchange and communication, because all radio links from a rocket to the ground (amateur TV, Wi-Fi connections) are managed by the

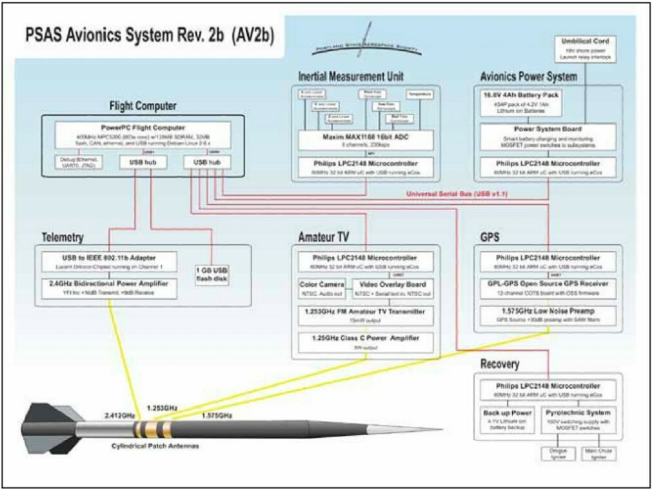

Figure 4: The avionics and flight control system is based on Linux